

Fall 25, 4100 Problem Set 9

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Sketch the conic section $x_1^2 + 8x_1x_2 + 7x_2^2 = 1$ by finding the spectral decomposition of

the matrix $A = \begin{bmatrix} 1 & 4 \\ 4 & 7 \end{bmatrix}$

$$x^T A x = 1$$

$$\det \begin{bmatrix} 1 - \lambda & 4 \\ 4 & 7 - \lambda \end{bmatrix} = (1 - \lambda)(7 - \lambda) - 16 = \lambda^2 - 8\lambda - 9 = (\lambda - 9)(\lambda + 1)$$

$$\lambda_1 = 9, \lambda_2 = -1, \text{ hyperbola}$$

$$(A - 9I)v_1 = \begin{bmatrix} -8 & 4 \\ 4 & -2 \end{bmatrix} v_1 = 0$$

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$(A + 1I)v_2 = \begin{bmatrix} 2 & 4 \\ 4 & 8 \end{bmatrix} v_2 = 0$$

$$v_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$u_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$u_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$Q = [u_1 \ u_2] = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

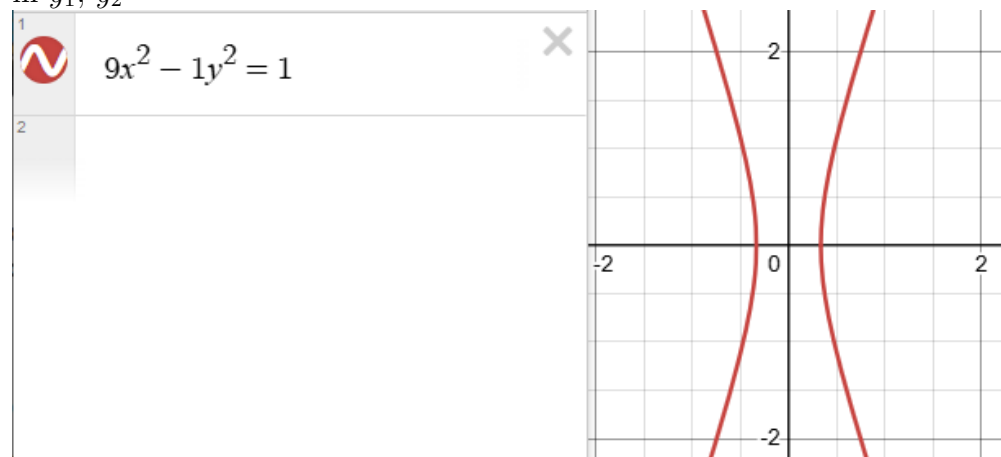
$$A = Q \Lambda Q^T$$

$$\Lambda = \begin{bmatrix} 9 & 0 \\ 0 & -1 \end{bmatrix}$$

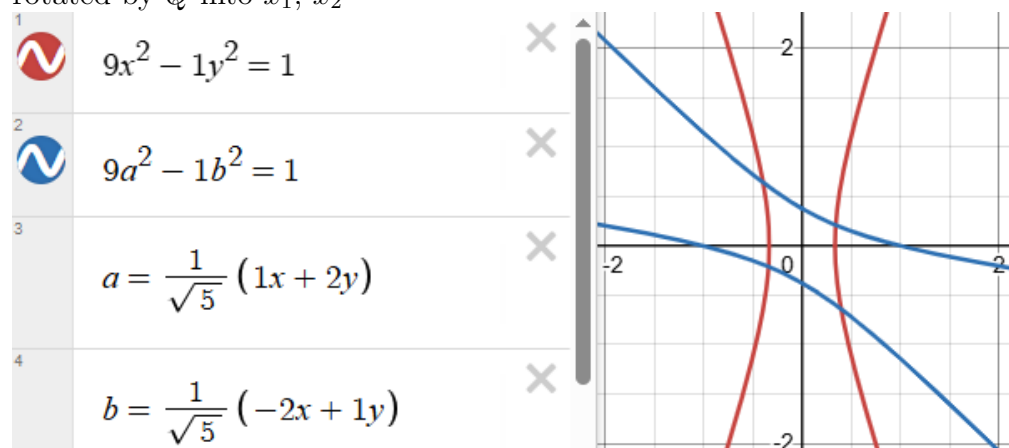
$$y = Q^T x, x^T A x = y^T \Lambda y$$

$$9y_1^2 - 1y_2^2 = 1$$

$$\text{in } y_1, y_2$$

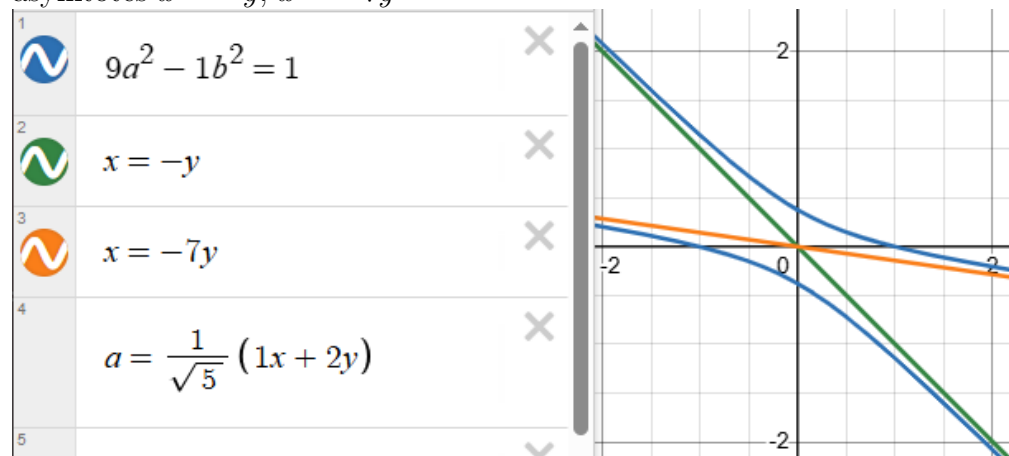


rotated by Q into x_1, x_2



$$(x + y)(x + 7y) = 1$$

asymptotes $x = -y, x = -7y$



2. Let A be any real matrix and prove that

- a. If $\mathbf{v}$ is an eigenvector of AA^T , then $\mathbf{v} \in \text{range}(A) \cup \text{null}(A^T)$ (look at the cases where the corresponding e-value is 0 or nonzero)

let A be an $m \times n$ matrix

$$AA^T \mathbf{v} = \lambda \mathbf{v}$$

first case: $\lambda = 0$

$$\text{then } AA^T \mathbf{v} = 0$$

$$\text{note } 0 = \mathbf{v}^T(AA^T \mathbf{v}) = \|A^T \mathbf{v}\|^2$$

$$\text{so } A^T \mathbf{v} = 0$$

$$\mathbf{v} \in \ker(A^T)$$

second case: $\lambda \neq 0$

$$\mathbf{v} = \frac{1}{\lambda} AA^T \mathbf{v} = A\left(\frac{1}{\lambda} A^T \mathbf{v}\right)$$

$$\text{so } \mathbf{v} \in \text{range}(A)$$

- b. If λ is a nonzero eigenvalue of AA^T , then it is also an eigenvalue of $A^T A$

$$\text{let } \lambda \neq 0 \text{ and } \mathbf{v} \neq 0, AA^T \mathbf{v} = \lambda \mathbf{v}$$

$$\text{set } \mathbf{u} := A^T \mathbf{v}$$

if $\mathbf{u} = 0$, then $A^T \mathbf{v} = 0$ would imply $AA^T \mathbf{v} = 0$, contradicting $\lambda \neq 0$, so $\mathbf{u} \neq 0$

$$A^T A \mathbf{u} = A^T A(A^T \mathbf{v}) = A^T(AA^T \mathbf{v}) = A^T(\lambda \mathbf{v}) = \lambda(A^T \mathbf{v}) = \lambda \mathbf{u}$$

hence $\mathbf{u}$ is an eigenvector of $A^T A$ with eigenvalue λ .

therefore any nonzero eigenvalue of AA^T is also an eigenvalue of $A^T A$

3. Let

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 1 & 2 \end{bmatrix}$$

a. Find the singular value decomposition of A

$$A^T A = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}$$

$$\lambda_1 = 7, \lambda_2 = 2$$

$$\sigma_1 = \sqrt{7}, \sigma_2 = \sqrt{2}$$

$$v_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, v_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$V = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

$$\text{let } u_i = \frac{1}{\sigma_i} A v_i$$

$$u_1 = \frac{1}{\sqrt{35}} \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}, u_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix},$$

$$\text{choose } u_3 = \frac{1}{\sqrt{14}} \begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix} \text{ orthonormal to } u_1, u_2$$

$$\Sigma = \begin{bmatrix} \sqrt{7} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{bmatrix}$$

$$A = U \Sigma V^T$$

b. Find the psuedo-inverse, A^+ and use it to find the vector in range A that is closest

$$\text{to } \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$A^+ = V \Sigma^+ U^T$$

$$\Sigma^+ = \begin{bmatrix} \sqrt{7} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{bmatrix}$$

$$A^+ = \begin{bmatrix} \frac{2}{7} & -\frac{4}{7} & \frac{1}{7} \\ \frac{1}{14} & \frac{5}{14} & \frac{2}{7} \end{bmatrix}$$

for any $b \in \mathbb{R}^3$ the orthogonal projection of b onto $\text{range}(A)$ is $P_{\text{range } A} b = A A^+ b$

$$\hat{x} = A^+ b = \begin{bmatrix} -\frac{3}{7} \\ \frac{23}{14} \end{bmatrix}$$

$$A \hat{x} = \begin{bmatrix} \frac{17}{14} \\ \frac{29}{14} \\ \frac{40}{14} \end{bmatrix}$$

4. Find and sketch the image of the unit circle under the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ where

$$T\mathbf{e}_1 = 2\mathbf{e}_1 + \mathbf{e}_2$$

$$T\mathbf{e}_2 = -3\mathbf{e}_1 + 6\mathbf{e}_2$$

using the method from lecture 11-20

$$A = \begin{bmatrix} 2 & -3 \\ 1 & 6 \end{bmatrix}$$

$$y = A^{-1}x, y^T y = 1, x^T (AA^T)^{-1} x = 1$$

$$AA^T = \begin{bmatrix} 13 & -16 \\ -16 & 37 \end{bmatrix}$$

$$(AA^T)^{-1} = \frac{1}{225} \begin{bmatrix} 37 & -16 \\ -16 & 13 \end{bmatrix}$$

$$\frac{37}{225}x_1^2 + \frac{32}{225}x_1x_2 + \frac{13}{225}x_2^2 = 1$$

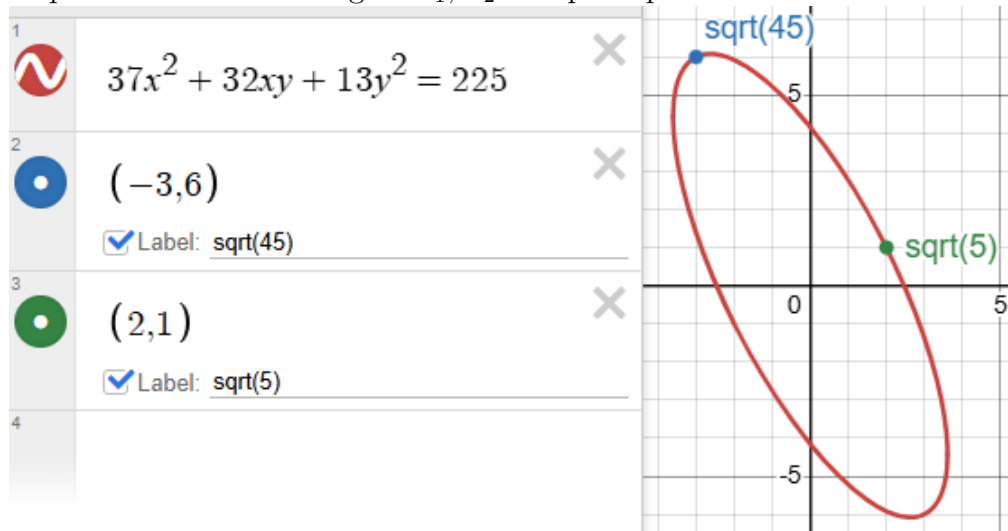
$$\lambda_1 = 25, \lambda_2 = 5$$

$$u_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} -\frac{1}{2} \\ 1 \end{bmatrix}$$

$$u_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\sigma_1 = 3\sqrt{5}, \sigma_2 = \sqrt{5}$$

ellipse with semi axis lengths σ_1, σ_2 and principle directions are columns u_1, u_2



5. Given $n \times n$ matrix A , define the trace of A as

$$\text{tr}(A) = \sum_{i=1}^n A_{i,i} = \langle \mathbf{e}_1, A\mathbf{e}_1 \rangle + \cdots + \langle \mathbf{e}_n, A\mathbf{e}_n \rangle$$

Now letting $A \in \mathbb{C}^{n \times n}$ have singular values of $\sigma_1, \dots, \sigma_r$, show that

$$\text{tr}(A^*A) = \sigma_1^2 + \cdots + \sigma_r^2$$

write SVD of A : $A = U\Sigma V^*$

where U, V are unitary and $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0)$

$$A^*A = (U\Sigma V^*)^*(U\Sigma V^*) = V\Sigma^*U^*U\Sigma V^* = V\Sigma^2V^*$$

$$U^*U = I \text{ and } \Sigma^* = \Sigma$$

$$\text{tr}(A^*A) = \text{tr}(V\Sigma^2V^*) = \text{tr}(V^*V\Sigma^2) = \text{tr}(\Sigma^2)$$

$$\Sigma^2 = \text{diag}(\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0)$$

$$\text{so } \text{tr}(\Sigma^2) = \sigma_1^2 + \cdots + \sigma_r^2$$

$$\text{Hence } \text{tr}(A^*A) = \sigma_1^2 + \cdots + \sigma_r^2$$